Supplementary Table S1 Quality scoring for included cohort studies using New castle-Ottawa Scale (NOS)

First author, year [ref.]	Selection				Comparability		Outcome			Total scores
	Representativeness of the exposed cohort	Selection of the non-exposed cohort	Ascertainment of exposure	Outcome of interest was not present at start of study	On basis of the design	On basis of the analysis	Assessment of outcome	Was follow-up long enough for outcomes to occur?	Adequacy of follow-up of cohorts	
Castón 2017	1	1	1	1	1	0	1	0	0	6
Shields 2018	1	1	1	1	1	0	1	0	0	6
Tumbarello 2019	0	1	1	1	1	0	1	1	1	7
Tsolaki 2019	1	1	1	1	1	0	1	1	0	7
Falcone 2021	1	1	1	1	1	0	1	1	1	8
Hakeam 2021	1	1	1	1	1	0	1	1	0	7

Supplementary Table S2 Quality scoring for included case-control study using New castle-Ottawa Scale (NOS)

First author, year [ref.]	Selection				Comparability		Exposure			Total score
	Case definition	Case representation	Control selection	Control definition	On basis of the design	On basis of the analysis	Assessment of exposure	Same method of ascertainment	Response rate	
Falcone 2020	1	1	1	0	1	0	1	1	1	7
Karaiskos 2020	1	1	1	1	1	1	1	1	0	8
Shen 2021	1	1	1	0	1	1	1	1	0	7
Zhou 2021	1	1	1	1	1	1	1	1	0	8
Chen 2021	1	1	1	0	1	1	1	1	0	7

Supplementary Table S3 Detailed antimicrobial therapy of patients with bloodstream infections caused by CRE

Reference	CAZ-AVI containing-regimen (n, %)	Other regimens (n, %)
Caston 2017	Monotherapy (2/8,25) Combination therapy (6/8,75): AG (7,87.5), CB (3,37.5), FOS (2,25), TIG (2,25), COL (2,25)	Monotherapy (11/23,47.8) Combination therapy (12/23,52.2)
Shields 2017	Monotherapy (8/13,61.5) Combination therapy (5/13,38.4): CAZ-AVI+GM (5,100)	Monotherapy (29/96,30.2): AG (11,37.9), CB (8,27.6), COL (4,13.8), TIG (4,13.8), ciprofloxacin (2,6.9) Combination therapy (67/96,69.8): CB+AG (25,37.3), CB+COL (30,44.8), COL+TIG (3,4.4), AG+TIG (2,3), AG+cefepime (1,1.5), AG+COL+TIG (1,1.5), COL+ATM (1,1.5), COL+cefepime (1,1.5), COL+ciprofloxacin (1,1.5), CB+doxycycline (1,1.5), CB+TIG (1,1.5)
Tumbarello 2019	Monotherapy (22/104,21.2) Combination therapy (82/104,78.8): CAZ-AVI+AK (2,2.4), CAZ-AVI+FOS (7,8.5), CAZ-AVI+TIG (16,19.5), CAZ-AVI+CB (19,23.2), CAZ-AVI+COL (13,15.9), CAZ-AVI+GM (25,30.5)	Monotherapy (27/104, 26): GM (14, 51.9), COL (9, 33.3), Others (4, 14.8) Combination therapy (77/104,74): Double CB (29,37.7), FOS+AK (13,16.9), FOS+GM (11,14.3), GM+CB (11,14.3), COL+FOS (10,13), Others (3,3.9)
Tsolaki 2019	Monotherapy (7/22,31.8)	Monotherapy (1/28,3.6)

	Combination therapy (15/22,68.2): AG (5,33.3), COL (12,80), TIG (8,53.3), FOS (3,20)	Combination therapy (27/28,96.4): AG (9,33.3), COL (24,88.8), TIG (23,85.2), TMP/SMX (2,7.4)
Falcone 2020	CAZ-AVI ± AG/FOS (13/13,100)	Monotherapy (0/78,0) Combination therapy (78/78,100): COL+CB+FOS/AG (61,78.2), TIG+CB (8,10.3), TIG+GM (6,7.7), Double CB (3,3.8)
Karaiskos 2020	CAZ-AVI-containing (71/71,100)	Monotherapy (29/71,40.8): AG (6,20.7), COL (9,31), TIG (14,48.3) Combination therapy (42/71,59.2)
Falcone 2021	Monotherapy (0/52,0) Combination therapy (52/52,100): CAZ-AVI+ATM (52,100)	Monotherapy (4/50,8): COL (2,50), ATM (2,50) Combination therapy (46/50,92): COL+FOS+TIG (7,15.2), COL+FOS (7,15.2), COL+CB (5,10.9), COL+ATM+BLIBL (4,8.7), COL+GM (1,2.2), COL+ cotrimoxazole (1,2.2), TIG+AG (8,17.4), FOS+AG (5,10.9), TIG+FOS (2,4.3), TIG+CB (1,2.2), ATM+AG (4,8.7), ATM+FOS (1,2.2)
Shen 2021	Monotherapy (4/9, 44.4)	TIG-containing (35/80,43.8), COL-containing (20/80,25), Others (25,31.3)

	Combination therapy (5/9, 55.6): CAZ-AVI+TIG (3,60), CAZ-AVI+COL (1,20), CAZ-AVI+CB (1,20)	
Zhou 2021	Monotherapy (1/4,25) Combination therapy (3/4,75): CAZ-AVI+CB (1,33.3), CAZ-AVI+TIG (2,66.7)	Monotherapy (91/131,69.5): TIG (71,78), CB (10,11), AK (6,6.6), COL (3,3.3), Other (1,1.1) Combination therapy (40/131, 30.5): TIG+COL (21,52.5), TIG+CB (6,15), TIG+AK (6,15), TIG+COL+CB (2,5), TIG+COL+AK (2,5), TIG+CB+AK (2,5), CB+AK (1,2.5)
Hakeam 2021	Monotherapy (9/32,28.1) Combination therapy (23/32,71.9): CB (4,17.4), ATM (7,30.4), Cefepime (1,4.3), AG (3,13), Fluoroquinolones (1,4.3), TIG (8,34.8), Vancomycin (7,30.4)	Monotherapy (0/29,0) Combination therapy (29/29,100): COL(29,100), CB (21,72.4), ATM (1,3.4), Cefepime (2,6.9), BLIBL (2,6.9), AG (6,20.7), Fluoroquinolones (1,3.4), TIG (9,31), Vancomycin (9,31)
Chen 2021	Monotherapy (13/35,37.1) Combination therapy (22/35,62.9): CAZ-AVI+TIG (13,59.1), CAZ-AVI+TIG+COL (9,40.9)	Monotherapy (19/152,12.5): TIG (19,100) Combination therapy (133/152,87.5): TIG+COL (46,34.6), CB+TIG+COL (44,33.1), CB+COL+AG (13,9.8), CB+TIG (14,10.5), CB+AG (16,12)

AG aminoglycosides, AK amikacin, ATM aztreonam, BLIBL Beta-lactamase-inhibiting beta-lactams, CB carbapenems, COL colistin, FOS fosfomycin, GM gentamycin, TIG tigecycline, TMP/SMX Trimethoprim-Sulfamethox

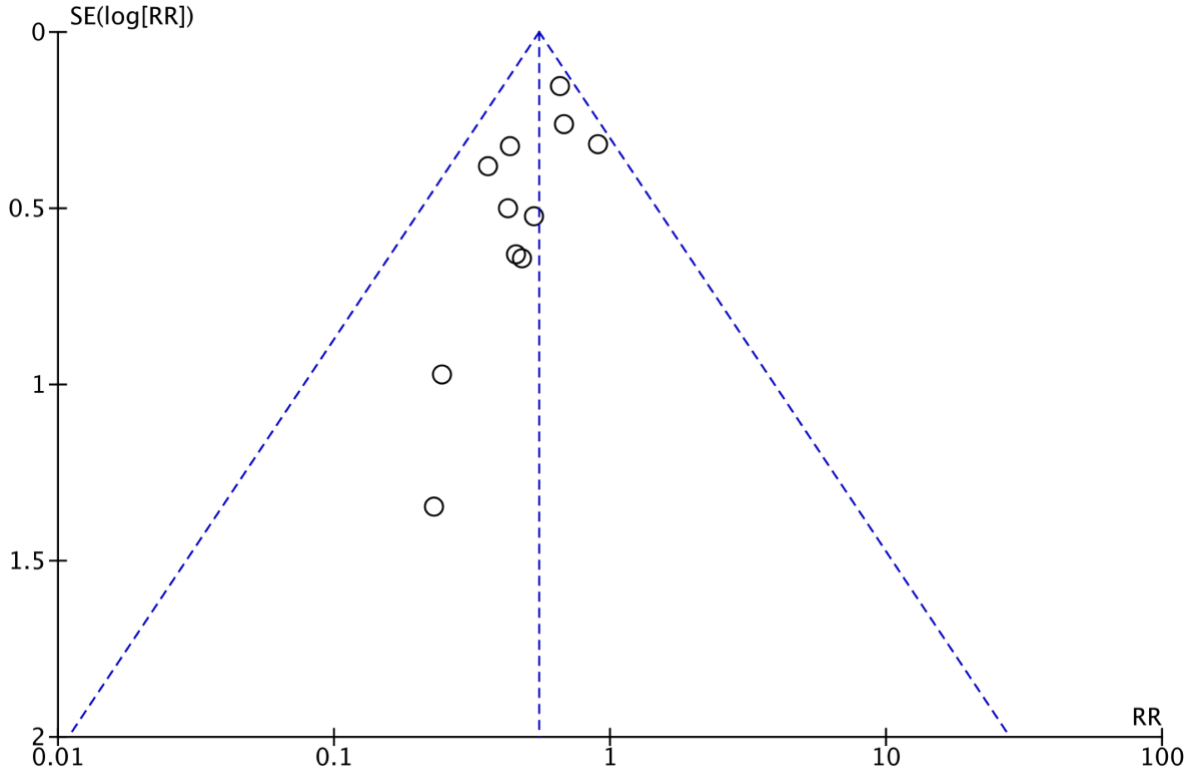

Figure S1. The funnel plot of publication bias

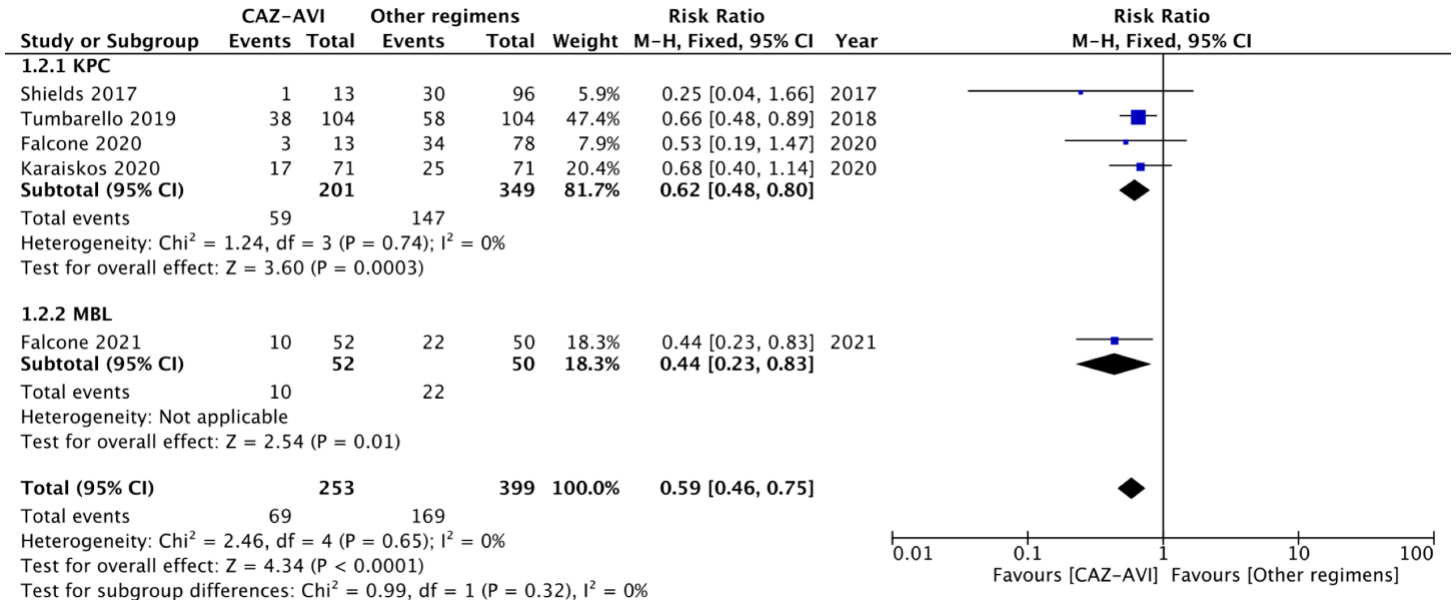

Figure S2. Subgroup analysis of carbapenemase on primary outcome in CRE BSI
MBL Metallo-β-lactamases

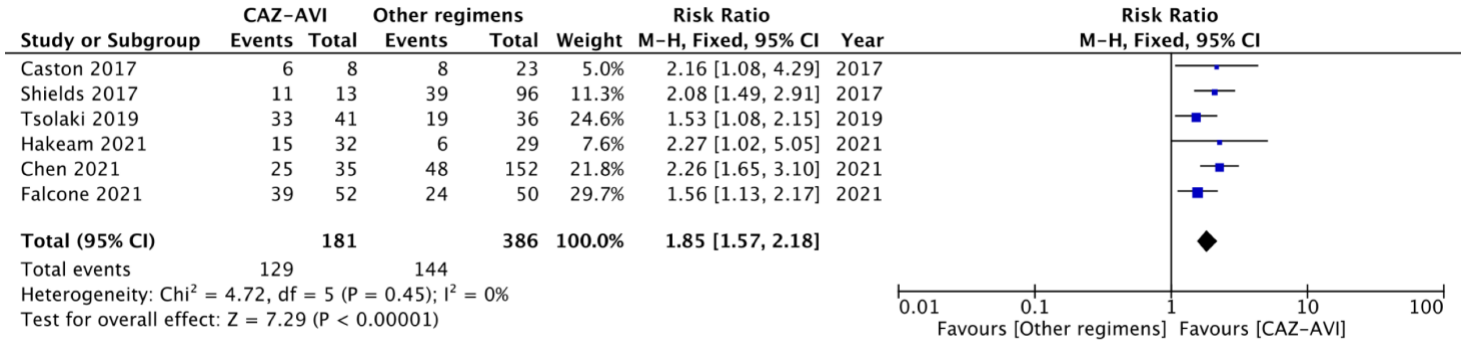

Figure S3. Clinical cure rate of CAZ-AVI regimens compared with controls in CRE BSI

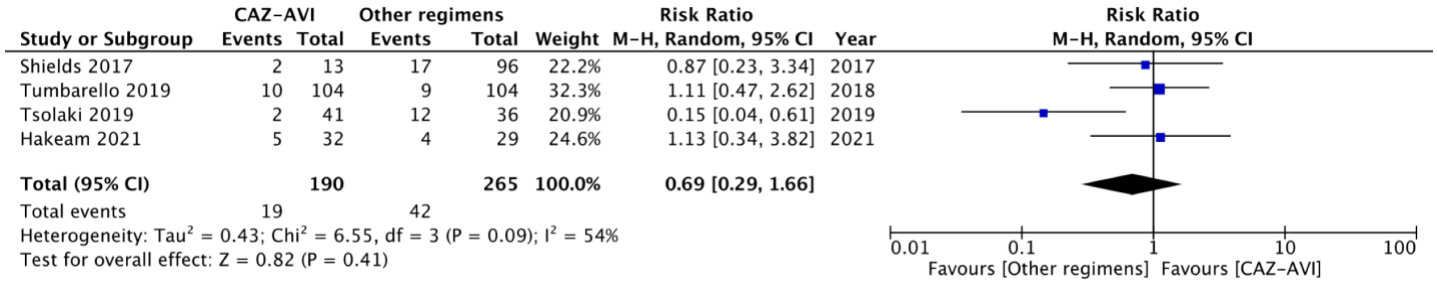

Figure S4. Relapse rate of CAZ-AVI regimens compared with controls in CRE BSI

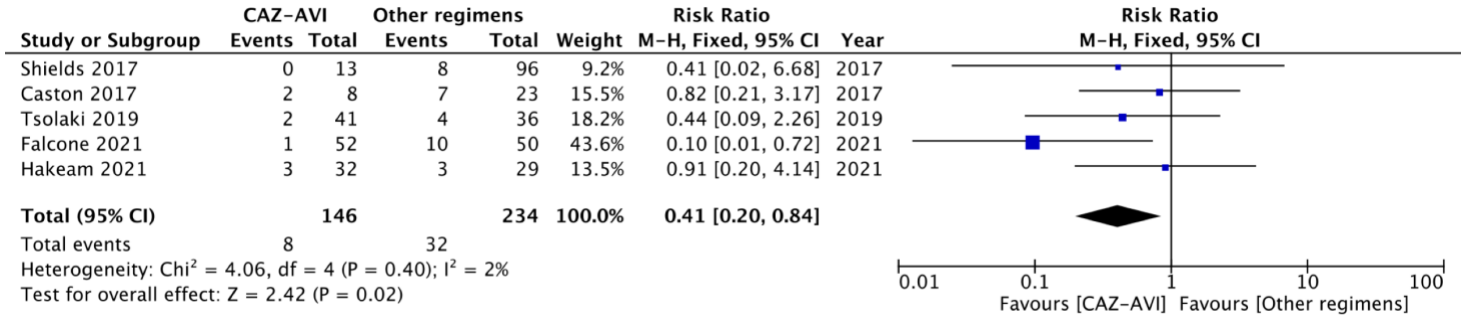

Figure S5. Nephrotoxicity of the CAZ-AVI regimens compared with control in CRE BSI

Search Strategy

Pubmed

Search ((((("avibactam, ceftazidime drug combination" [Supplementary Concept]) OR ((((((ceftazidime/avibactam[Title/Abstract]) OR ceftazidime-avibactam[Title/Abstract]) OR avibactam-ceftazidime[Title/Abstract]) OR Avycaz[Title/Abstract]) OR NXL104[Title/Abstract]) OR AVE1330A[Title/Abstract]))) AND ((((((((((((((carbapenem resist*[Title/Abstract]) OR carbapenemase producing[Title/Abstract]) OR carbapenem resistance[Title/Abstract]) OR KPC[Title/Abstract]) OR Metallo Lactamase[Title/Abstract]) OR Metallo beta lactamase[Title/Abstract]) OR VIM-producing[Title/Abstract]) OR VIM produc*[Title/Abstract]) OR IMP-producing[Title/Abstract]) OR IMP produc*[Title/Abstract]) OR NDM-producing[Title/Abstract]) OR NDM produc*[Title/Abstract]) OR OXA[Title/Abstract]) OR CRE[Title/Abstract]) OR Carbapenem-resistant[Title/Abstract])) AND ((risk[Title/Abstract] OR risk[MeSH:noexp] OR mortality[Title/Abstract] OR mortality[MeSH:noexp] OR cohort[Title/Abstract]))

Embase

#12.	#7 AND #10 AND #11	166	5 Nov 2021
#11.	'risk':ab,ti OR 'mortalit':ab,ti OR 'cohort':ab,ti	4,070,182	5 Nov 2021
#10.	#8 OR #9	62,797	5 Nov 2021
#9.	'carbapenem resist*':ab,ti OR 'carbapenemase producing':ab,ti OR 'carbapenem resistance':ab,ti OR 'kpc':ab,ti OR 'metallo lactamase':ab,ti OR 'metallo beta lactamase':ab,ti OR 'vim-producing':ab,ti OR 'vim produc*':ab,ti OR 'imp-producing':ab,ti OR 'imp produc*':ab,ti OR 'ndm-producing':ab,ti OR 'ndm produc*':ab,ti OR 'oxa':ab,ti OR 'cre':ab,ti OR 'carbapenem-resistant':ab,ti	62,227	5 Nov 2021
#8.	'carbapenem-resistant enterobacteriaceae'/exp	3,442	5 Nov 2021
#7.	#1 OR #2 OR #3 OR #4 OR #5 OR #6	1,801	5 Nov 2021

#6.	'ave1330a':ab,ti	6	5 Nov 2021
#5.	'nxl104':ab,ti	55	5 Nov 2021
#4.	'avycaz':ab,ti	10	5 Nov 2021
#3.	'ceftazidime avibactam':ab,ti	1,136	5 Nov 2021
#2.	'avibactam plus ceftazidime'/exp	1,593	5 Nov 2021
#1.	'avibactam ceftazidime':ab,ti	23	5 Nov 2021

Cochrane

#1 (avibactam ceftazidime):ti,ab,kw OR (Avycaz):ti,ab,kw OR (NXL104):ti,ab,kw OR (AVE1330A):ti,ab,kw OR (avibactam, ceftazidime drug combination):ti,ab,kw (Word variations have been searched) 91

#2 MeSH descriptor: [Carbapenem-Resistant Enterobacteriaceae] explode all trees 4

#3 (carbapenem resist*):ti,ab,kw OR (carbapenemase producing):ti,ab,kw OR (carbapenem resistance):ti,ab,kw OR (KPC):ti,ab,kw OR (Metallo Lactamase):ti,ab,kw (Word variations have been searched) 368

#4 (Metallo beta lactamase):ti,ab,kw OR (VIM-producing):ti,ab,kw OR (VIM produc*):ti,ab,kw OR (IMP-producing):ti,ab,kw OR (IMP produc*):ti,ab,kw (Word variations have been searched) 774

#5 (NDM-producing):ti,ab,kw OR (NDM produc*):ti,ab,kw OR (OXA):ti,ab,kw OR (CRE):ti,ab,kw OR (Carbapenem-resistant):ti,ab,kw (Word variations have been searched)1127

#6 #2 OR #3 OR #4 OR #5 2047

#7 #1 AND #6 24